IAT Test 1

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 In plant X, the conspicuous green generation produces gametes; the spore-producing generation remains attached to it and nutritionally dependent on it. In plant Y, the conspicuous green generation produces spores and is independent. Which identification is consistent with these observations?
A) X is a fern and Y is a moss
B) X is a moss and Y is a fern
C) Both X and Y must be mosses
D) Both X and Y must be ferns

Q2 Two young stem sections are examined. Section P has many scattered, conjoint vascular bundles without cambium between xylem and phloem. Section Q has conjoint bundles in a ring, with cambium between xylem and phloem. Under the usual NCERT model of stem anatomy, which prediction is best supported?
A) P should undergo ordinary cambium-driven secondary growth, but Q should not
B) Both sections are roots because xylem and phloem occur in the same bundle
C) Both sections have closed vascular bundles
D) Q can establish a vascular cambium ring for secondary growth; P normally cannot

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Q3 Three equal preparations of an enzyme are assayed at the same final pH and temperature. U is untreated. I contains a reversible inhibitor. H was heated strongly and then cooled. Approximate initial rates are: Low substrate: U = 40, I = 20, H = 4 units. Very high substrate: U = 100, I = 100, H = 10 units. Rates are rounded to the experimental precision. Which interpretation is consistent with both substrate conditions?
A) I is consistent with competitive inhibition; heating has reduced the amount of functional enzyme in H
B) I must have destroyed half the enzyme molecules irreversibly; H is fully restored by cooling
C) I has doubled the maximum catalytic rate; H has increased substrate affinity
D) Both I and H merely reduce substrate concentration, with no effect on enzyme function

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Q4 A test-cross of a double heterozygote produces 420 Ab, 420 aB, 80 AB and 80 ab progeny. Assume these frequencies accurately represent gamete frequencies, recombination is the same in male and female meiosis, and there is no viability selection. If two individuals with this same double-heterozygous genotype and phase are crossed, what percentage of their offspring is expected to be aabb?
A) 6.4%
B) 17.64%
C) 0.64%
D) 25%

Q5 Urine measurements before and after an intervention are shown. Glomerular filtration and the medullary osmotic gradient stay constant. Condition | Urine volume (L/day) | Osmolarity (mOsm/L) Before | 2.0 | 300 After | 0.75 | 800 Which interpretation best combines the measured solute output with the likely tubular response?
A) Solute output falls; collecting-duct water permeability decreases
B) Solute output is unchanged; collecting-duct water permeability increases
C) Solute output increases; collecting-duct water permeability increases
D) Solute output is unchanged; collecting-duct water permeability decreases

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Q6 An ecosystem's plants fix 2400 kJ m−2 day−1 and use 900 kJ m−2 day−1 in respiration. Herbivores assimilate 180 kJ m−2 day−1 of plant-derived energy and respire 120 kJ m−2 day−1; the remainder becomes new herbivore biomass. What percentage of plant net primary production becomes this new herbivore biomass?

A) 4%
B) 7.5%
C) 12%
D) 2.5%

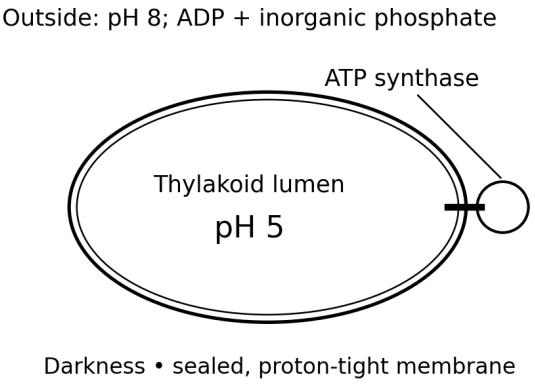
Q7 A diploid cell has 12 chromosomes in G1. It completes S phase and reaches G2 without mitosis. If its G1 DNA content is 2C, what are its chromosome number and DNA content in G2? Count chromosomes by centromeres, before sister chromatids separate.

A) 24 chromosomes and 4C DNA
B) 12 chromosomes and 2C DNA
C) 24 chromosomes and 2C DNA
D) 12 chromosomes and 4C DNA

Q8 A diploid female angiosperm has 2n = 24 chromosomes and a tetraploid male of the same species has 4n = 48. Assume normal meiosis, a typical embryo sac with two reduced polar nuclei, successful double fertilisation and no chromosome loss. What chromosome counts and maternal:paternal chromosome contributions are expected in the zygote and primary endosperm nucleus, respectively?

A) Zygote 24, 1:1; endosperm 36, 2:1
B) Zygote 36, 1:2; endosperm 72, 2:1
C) Zygote 36, 1:2; endosperm 48, 1:1
D) Zygote 48, 1:1; endosperm 36, 1:2

Q9 A sealed thylakoid preparation is placed in darkness with the imposed pH difference shown. Its membrane is otherwise proton-tight, its ATP synthase is functional, and ADP and inorganic phosphate are available outside. No other energy source is supplied. Which short-term observation is expected?

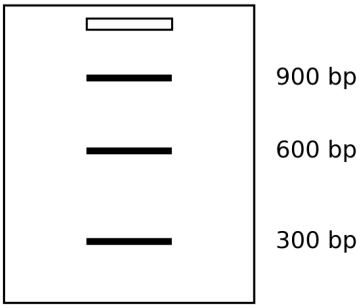


A) Transient ATP synthesis; no light-driven NADPH formation
B) Transient NADPH formation; no ATP synthesis from the gradient
C) Neither ATP nor NADPH formation before illumination
D) Both products form while protons enter the acidic lumen

Q10 In an immunology comparison, preparation P contains ready-made antitoxin antibodies, whereas preparation Q contains a harmless toxoid that elicits the recipient's own immune response. Which comparison is expected in an immunologically responsive recipient?
A) P and Q both supply memory cells and act immediately
B) P supplies prompt protection; Q can induce delayed immune memory
C) Q supplies prompt protection; P can induce delayed immune memory
D) P and Q both supply antibodies without inducing memory

Q11 A diploid locus is PCR-amplified to a 900 bp product. Allele R has one restriction site, giving 600 bp and 300 bp fragments after complete digestion; allele r has no site. The sample lane is shown. Assume complete digestion, equal PCR amplification per allele, equal DNA recovery, no other fragments and fluorescence proportional to DNA mass. What fraction of the sample-lane fluorescence should occur in the 900 bp band?

Completely digested PCR sample



Band positions only; intensity is not to scale

A) One-third
B) One-sixth
C) One-half
D) Two-thirds

Q12 Viable muscle-cell preparations A and B have equal basal glucose uptake. A reagent S activates an intracellular step downstream of insulin reception, without binding insulin receptors. Uptake measurements are: Treatment | A (units) | B (units) None | 2 | 2 Insulin | 8 | 2 S | 8 | 8 Insulin binding to the cell surface is normal in both preparations. Which conclusion is best supported?
A) B cannot take up glucose through any membrane pathway
B) B fails because its cells cannot synthesise insulin
C) B's insulin receptors cannot bind the supplied hormone
D) B has a defect after binding but before the step activated by S

Q13 A fully penetrant autosomal-recessive disorder is caused by allele a. An unaffected person has an affected sibling and two unaffected parents; assume the parents are Aa × Aa. This person has a child with an unrelated, unaffected partner from a Hardy–Weinberg population where the frequency of a is 0.20. Assume random mating, no new mutation and independent family origins. What is the probability that the child is affected?
A) 1/12
B) 1/18
C) 1/24
D) 1/6

Q14 Two resting frogs are fully submerged at the same temperature without lung ventilation. Their intact, moist skin has equal area and thickness, and skin blood flow is maintained. Frog | Dissolved oxygen in water (relative units) | Oxygen uptake (relative units) P | 8 | 4 Q | 4 | 2 Assume dissolved oxygen immediately inside the respiratory surface is negligible in both. Which interpretation and prediction best fit these observations?
A) Skin diffusion fits the data; lowering P's water oxygen to 4 should lower its uptake
B) Lung ventilation fits the data; raising Q's skin thickness should raise its uptake
C) Equal skin area requires equal uptake, regardless of the oxygen gradient
D) Uptake is independent of water oxygen if skin blood flow stays constant

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Q15 An initially neutral allele has frequency 0.50 in each of eight isolated experimental populations. Four populations are very small and four are very large. After many generations, the small populations have allele frequencies 0.00, 0.10, 0.90 and 1.00; the large populations remain near 0.50. Mutation, migration and fitness differences are excluded. Which explanation best fits the replicate outcomes?
A) Directional selection favours the allele equally in every population
B) The allele must have become beneficial in the two populations where it increased
C) Random sampling of alleles has stronger effects in the small populations
D) Hardy–Weinberg proportions require allele frequencies to change identically in all populations

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Chemistry

Q16 An isolated carbonate ion, CO_3^{2-}, is described using its three equivalent principal Lewis resonance structures. Which statement about its three carbon–oxygen bonds is correct?
A) Unequal bonds; one permanent double bond
B) Unequal bonds; switching double-bond location
C) Equivalent bonds; average bond order 2
D) Equivalent bonds; average bond order 4/3

Q17 Compare octahedral $[Co(NH_3)_6]^{3+}$ and $[Co(CN)_6]^{3-}$. Assume the observed absorption in each comparison is the transition associated with its octahedral splitting Δ_o. Cyanide produces the larger splitting. Which statement follows?
A) The cyanide complex absorbs at shorter wavelength, while cobalt is +3 in both complexes
B) The cyanide complex absorbs at longer wavelength, while cobalt is +3 in both complexes
C) The two complexes must absorb at the same wavelength because both have six ligands
D) The cobalt oxidation states are +3 and −3, respectively, so ligand strength is irrelevant

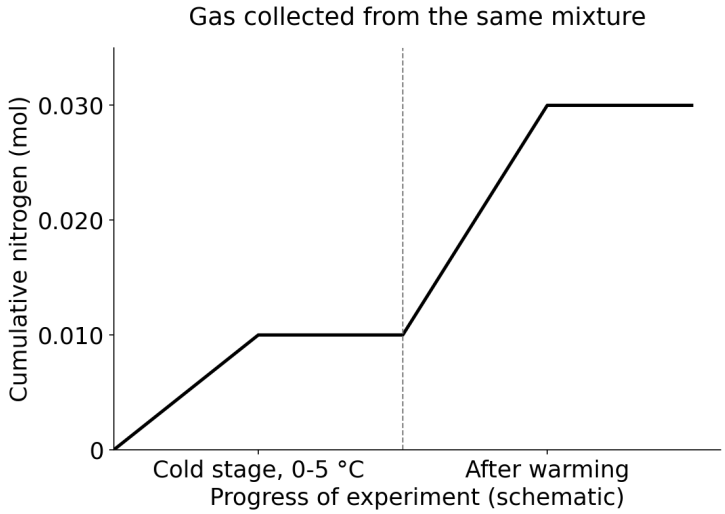
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Q18 For a reaction, $\Delta H = +24.0$ kJ mol^{-1} and $\Delta S = +80.0$ J K^{-1}mol^{-1}, independent of temperature over the range considered. Which pair gives ΔG at 200 K and 400 K, respectively?
A) −8.0 and +8.0 kJ mol−1
B) +40.0 and +56.0 kJ mol−1
C) +8.0 and −8.0 kJ mol−1
D) +23.984 and +23.968 kJ mol−1

Q19 A mixture contains only propanal and propanone. Separate complete tests on identical portions of the mixture give 2.16 g of silver with Tollens reagent and 3.94 g of iodoform in the iodoform test. Propanal gives no iodoform under these conditions. Take $M(\text{Ag}) = 108$ g mol^{-1} and $M(CHI_3) = 394$ g mol^{-1}. What is the mole fraction of propanone in the mixture?
A) 1/3
B) 1/2
C) 2/3
D) 1/4

Q20 A zinc–copper galvanic cell has $E^\circ_{\text{cell}} = 1.10$ V and $[Zn^{2+}] = 1.0$ M. In the copper half-cell, solid $Cu(OH)_2$ is present and a buffer maintains $[OH^-] = 10^{-4}$ M. Given $K_{sp}(Cu(OH)_2) = 10^{-20}$, estimate the cell emf. Treat activities as concentrations, neglect junction potentials, and take $2.303RT/F = 0.060$ V.
A) 1.46 V
B) 1.10 V
C) 0.38 V
D) 0.74 V

Q21 A mixture contains only methylamine and aniline. It is treated with excess nitrous acid at 0–5 °C, then warmed after cold-stage gas evolution stops. The graph shows the cumulative nitrogen collected. Assume quantitative reactions, negligible gas loss and complete thermal decomposition of the aromatic diazonium salt on warming. Which initial amounts of methylamine and aniline, respectively, are consistent with the graph?



A) 0.030 mol; 0.010 mol
B) 0.020 mol; 0.010 mol
C) 0.010 mol; 0.020 mol
D) 0.010 mol; 0.030 mol

Q22 Treat Fe_3O_4 as containing only Fe^{2+}, Fe^{3+} and O^{2-}. What fraction of its iron ions must be in the +3 oxidation state?
A) 2/3
B) 1/3
C) 1/2
D) 1

Q23 A first-order reaction is 25% complete after 10.0 min. How much additional time is required for the reaction to become 75% complete? Take $\ln(4/3) = 0.288$ and $\ln 3 = 1.099$.
A) 20.0 min
B) 38.2 min
C) 48.2 min
D) 30.0 min

Q24 Racemic $CH_3COCH(CH_3)CH{=}CH_2$ is treated with $NaBH_4$, reducing only the ketone to an alcohol. Both faces of the carbonyl can react. After work-up, all the alcohol is acylated with ethanoyl chloride in pyridine. The terminal C=C bond survives, and no stereocentre subsequently changes configuration. What are the molecular formula and maximum number of distinct configurational stereoisomers of the final ester? Count all products accessible from both starting enantiomers; ignore conformers.
A) $C_8H_{12}O_2$; 4
B) $C_8H_{14}O_2$; 2
C) $C_8H_{14}O_2$; 4
D) $C_8H_{14}O_2$; 8

Q25 The figure gives the connectivity of alkenes P, Q and R; it does not assign a particular geometrical configuration to Q. Which classification is correct? Ignore conformers and count geometrical isomerism only where each double-bond carbon has two different substituents.

Alkene connectivity

P $CH_2=CH\text{-}CH_2\text{-}CH_3$

Q $CH_3\text{-}CH=CH\text{-}CH_3$

R $CH_2=C(CH_3)_2$

Connectivity only; no E/Z configuration assigned

A) P and Q are chain isomers; P and R are position isomers; all three show E/Z isomerism
B) P and Q are position isomers; P and R are chain isomers; P and Q show E/Z isomerism
C) P and Q are chain isomers; P and R are position isomers; only R shows E/Z isomerism
D) P and Q are position isomers; P and R are chain isomers; only Q shows E/Z isomerism

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Q26 Two solutions at temperature T initially have equal hydrostatic pressure and are separated by an ideal membrane permeable only to water. The left side contains 0.10 M sucrose. The right side contains 0.060 M $CaCl_2$, of which 75% is dissociated; assume ideal behaviour otherwise. Which initial water-flow direction and extra pressure would prevent that flow? Use R in L atm mol−1 K−1.
A) Water flows right to left; apply an extra 0.040RT atm on the left
B) Water flows left to right; apply an extra 0.050RT atm on the right
C) Water flows left to right; apply an extra 0.080RT atm on the right
D) No initial flow occurs because both solutes have the same mass concentration

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Q27 A 1.90 g mixture contains only anhydrous Na_2CO_3 and $NaHCO_3$. Heating completely decomposes the bicarbonate and leaves 1.59 g of anhydrous residue. What was the mass percentage of sodium bicarbonate in the original mixture? Use molar masses 106 and 84 g/mol for sodium carbonate and sodium bicarbonate, respectively; the only volatile products are carbon dioxide and water.
A) 44.2%
B) 22.1%
C) 55.8%
D) 16.3%

Q28 Consider the ions S^{2-}, Cl^-, K^+ and Ca^{2+}. Their atomic numbers are 16, 17, 19 and 20, respectively. Which order of ionic radii is expected for these isoelectronic ions?
A) $Ca^{2+} > K^+ > Cl^- > S^{2-}$
B) $K^+ > Ca^{2+} > S^{2-} > Cl^-$
C) $Cl^- > S^{2-} > Ca^{2+} > K^+$
D) $S^{2-} > Cl^- > K^+ > Ca^{2+}$

Q29 Separate carbohydrate samples X, Y and Z give the results below. Hydrolysates are neutralised before testing. Positive results mean a reducing sugar is present; hydrolysis is complete. Sample | Reducing-sugar test before hydrolysis | Products after hydrolysis X | Negative | Glucose and fructose Y | Positive | Glucose only Z | Positive | Glucose and galactose Which identification fits all observations?
A) X maltose; Y sucrose; Z lactose
B) X sucrose; Y lactose; Z maltose
C) X sucrose; Y maltose; Z lactose
D) X lactose; Y maltose; Z sucrose

Q30 100 mL of 0.40 M ammonia is mixed with 100 mL of 0.10 M HCl. Then 0.020 mol more HCl is added in a small volume. Assuming quantitative neutralisation followed by weak-base equilibrium, what is the final pH? Take $K_b(\mathrm{NH_3}) = 10^{-5}$, $pK_w = 14.00$ and $\log 3 = 0.477$. Activity effects may be neglected.
A) 8.52
B) 9.48
C) 5.00
D) 7.00

Mathematics

Q31 For $A = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$, what is the sum of the diagonal entries of A^{-1}?

A) 4
B) 1/2
C) 2
D) −2

Q32 Six distinct folders A, B, C, D, E and F are arranged in a row. A must be to the left of B, with exactly one folder between A and B. How many arrangements satisfy these conditions?

A) 48
B) 96
C) 192
D) 120

Q33 A function $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x) + 2f(1 - x) = x^2$ for every real x. What is the minimum value of f on $[-1, 2]$?

A) 0
B) 2/3
C) −1/3
D) −2/3

Q34 Two complex numbers z_1 and z_2 are exactly the solutions of $|z - 1| = |z + 1|$ and $|z - 2i| = \sqrt{5}$. What is $\mathrm{Im}(z_1)\,\mathrm{Im}(z_2)$?

A) −1
B) 1
C) −5
D) 4

Q35 Evaluate $I = \displaystyle\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\, dx$.

A) π/2
B) π²/2
C) π²/4
D) π²/8

Q36 Point P has coordinates (1,2,3). Its reflection in the plane $x + y + z = 3$ is P′. What is the distance of P′ from the origin?

A) √14
B) √5
C) 2√3
D) √2

Q37 Real numbers a and b are chosen so that $f(x) = |x^2 - 1| + a|x - 1| + b|x + 1|$ is differentiable at every real x. What is a+b?

A) 4
B) −4
C) −2
D) 0

Q38 Two groups each contain four observations. Group A has mean 2 and population variance 1; group B has mean 6 and population variance 9. Here each variance uses the number of observations, not one less, as its denominator. What is the population variance of all eight observations together?
A) 9
B) 5
C) 13
D) 18

Q39 Find $\lim_{x \to 0} \dfrac{\sqrt{1+3x} - \sqrt{1-x}}{x}$.
A) 4
B) 1
C) 0
D) 2

Q40 Six distinct keys are assigned randomly, one each, to six numbered locks. Exactly one key is correct for each lock, and every one of the $6!$ assignments is equally likely. What is the probability that exactly two locks receive their correct keys?
A) 1/8
B) 1/48
C) 3/16
D) 1/2

Q41 Using principal values of inverse tangent, evaluate $\tan^{-1} 2 + \tan^{-1} 3$.
A) $3\pi/4$
B) $-\pi/4$
C) $\pi/4$
D) $\tan^{-1} 5$

Q42 For $x > 0$, a function satisfies $\dfrac{dy}{dx} = \dfrac{y}{x(1+x)}$ and y(1)=2. What is y(3)?
A) 3/2
B) 3
C) 6
D) 4

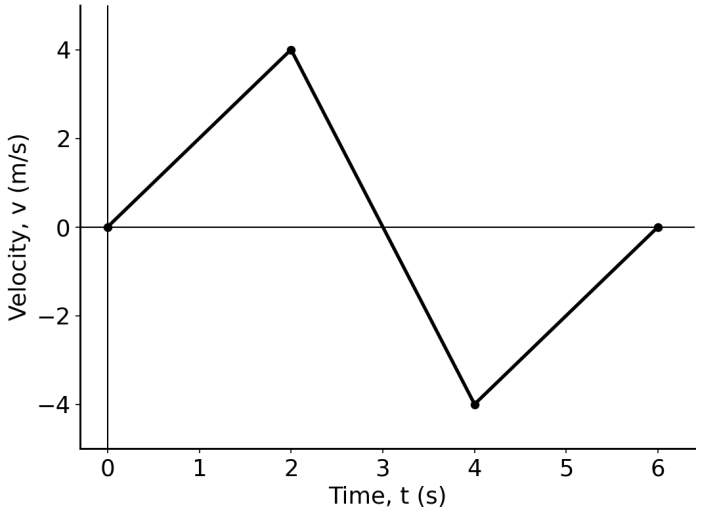
Q43 For which real values of a does $x^2 - 2ax + a + 2 = 0$ have two real roots r1 and r2 satisfying $r_1 < 1 < r_2$?
A) a ≥ 3
B) a < −1 or a > 2
C) a > 3
D) a < 3

Q44 The line of slope 1 through the focus of the parabola $y^2 = 4x$ meets it at P and Q. The tangents to the parabola at P and Q meet at T. What is the area of triangle PQT, in square units?
A) $4\sqrt{2}$
B) 8
C) $16\sqrt{2}$
D) $8\sqrt{2}$

Q45 Unit vectors **a** and **b** make an angle of 60°. A parallelogram has adjacent side vectors $\mathbf{a}+2\mathbf{b}$ and $\mathbf{a}-\mathbf{b}$. What is its area?
A) √3/2
B) 3√3/2
C) 3
D) 3√3/4

Physics

Q46 A particle moves along a straight line with the velocity-time graph shown. What are its average speed and average velocity, respectively, over the full 6 s interval?



A) $2\,\mathrm{m\,s^{-1}},\ 0$
B) $0,\ 2\,\mathrm{m\,s^{-1}}$
C) $4/3\,\mathrm{m\,s^{-1}},\ 0$
D) $2\,\mathrm{m\,s^{-1}},\ 2\,\mathrm{m\,s^{-1}}$

Q47 An empirical pressure model has the form $P = a\rho v^2 + bh$, where density is ρ, speed is v, height is h, and a is dimensionless. What are the dimensions of b?
A) $ML^{-1}T^{-2}$
B) LT^{-2}
C) $ML^{-2}T^{-2}$
D) ML^{-3}

Q48 An incompressible, nonviscous fluid of density ρ flows steadily through a horizontal pipe. A wide section has radius r and speed v; a narrow section has radius r/2. What is the pressure in the wide section minus that in the narrow section?
A) $\frac{3}{2}\rho v^2$
B) $\frac{15}{2}\rho v^2$
C) $8\rho v^2$
D) $4\rho v^2$

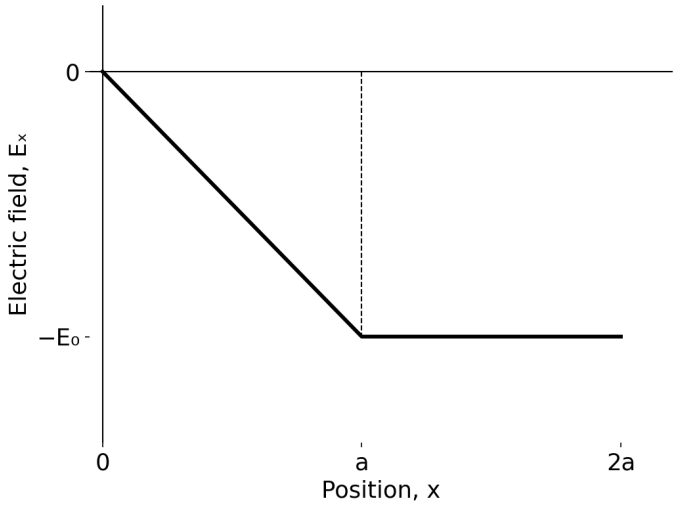
Q49 A horizontal disc has constant moment of inertia $2mR^2$ about its frictionless vertical axle. A point bead of mass m initially at radius R co-rotates with it at angular speed ω. A powered actuator attached to the disc moves the bead slowly along a radial groove to R/2. Initially and finally the bead has no radial velocity and co-rotates with the disc. The actuator has negligible inertia; there is no external torque about the axle and no mechanical dissipation. What are the final angular speed and the work supplied by the actuator to the disc-bead mechanical motion?
A) $3\omega/4,\ -mR^2\omega^2/2$
B) $4\omega/3,\ 0$
C) $\omega,\ mR^2\omega^2/2$
D) $4\omega/3,\ mR^2\omega^2/2$

Q50 A cyclic heat engine absorbs 800 J from a reservoir at 500 K, delivers 300 J of work and rejects the remaining heat to a reservoir at 300 K. Which statement is correct?
A) These figures are consistent with an irreversible engine.
B) These figures describe a reversible engine.
C) These figures necessarily violate the first law of thermodynamics.
D) Its efficiency exceeds the Carnot limit for these reservoirs.

Q51 A mass m stretches an ideal massless vertical spring by 10 cm at equilibrium. Let $\omega = \sqrt{k/m}$. The mass is initially 3 cm below equilibrium and is given a downward speed $v_0 = (0.04\text{ m})\omega$. With no damping, what is the maximum tensile force in the spring?
A) mg/2
B) 3mg/2
C) 17mg/10
D) mg

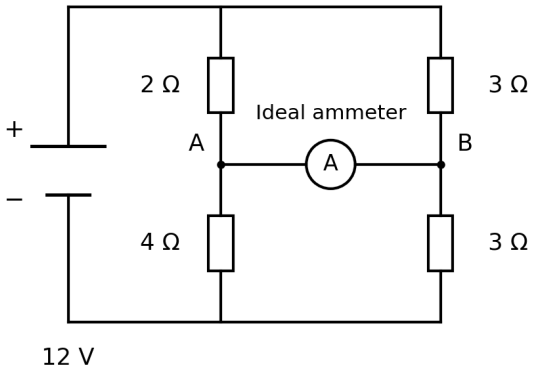
Q52 An unknown tuning fork produces 4 beats per second with the fundamental of a stretched string at 512 Hz and tension T. The string's vibrating length and linear mass density remain constant. Reducing its tension to $(255/256)^2T$ makes the beat frequency 2 Hz. What is the fork frequency, and what tension would make the beats vanish?
A) 506 Hz; $(253/256)^2T$
B) 510 Hz; $(255/256)^2T$
C) 508 Hz; $(127/128)^2T$
D) 516 Hz; $(129/128)^2T$

Q53 A positive charge q moves along the x axis under only the electric force. The field component is shown; $E_0 > 0$ and $a > 0$. At x = 0 the charge moves towards increasing x with kinetic energy $5qE_0a/4$. At what coordinate does it first come to rest?



A) 5a/4
B) 3a/2
C) 2a
D) 7a/4

Q54 In the circuit shown, the top rail is 12 V above the bottom rail. The source and ammeter are ideal; all other wires have negligible resistance. What current flows through the ammeter?

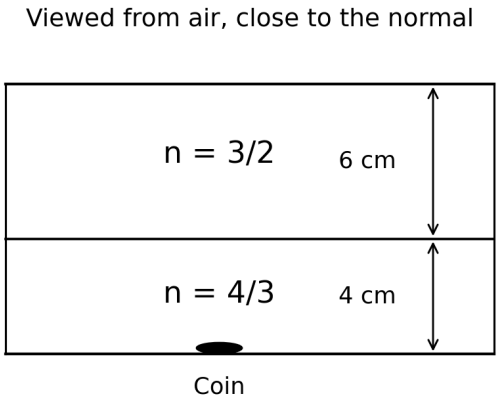


A) 2 A, from A to B
B) $12/17$ A, from A to B
C) $12/17$ A, from B to A
D) Zero

Q55 A nucleus initially has mass number 238 and atomic number 92. It undergoes one alpha decay followed by two beta-minus decays, with no other nuclear transformation. What are its final mass number and atomic number, respectively?
A) 234, 92
B) 234, 90
C) 238, 94
D) 236, 92

Q56 A conducting bar of mass m and length l spans two parallel conducting rails perpendicular to their length and slides down them. The rails are inclined at angle θ to the horizontal. A uniform magnetic field of magnitude B is perpendicular to the rail plane. The closed circuit has constant total resistance R. Neglect friction, inductance and all other drag. What is the terminal speed of the bar?
A) $mgR\cos\theta/(B^2l^2)$
B) $mgR\sin\theta/(2B^2l^2)$
C) $2mgR\sin\theta/(B^2l^2)$
D) $mgR\sin\theta/(B^2l^2)$

Q57 A coin lies at the bottom of the two transparent, immiscible liquid layers shown. It is viewed from air close to the normal. Neglect container thickness. What is its apparent depth below the top liquid surface, and would reversing the order of the layers (keeping each layer's depth) change this apparent depth?



A) 10 cm; unchanged
B) 7 cm; changes
C) 7 cm; unchanged
D) $43/3$ cm; unchanged

Q58 A series circuit contains an ideal resistor, inductor and capacitor driven by a sinusoidal source. The rms current is 2 A, and the rms voltage magnitudes across R, L and C are 8 V, 10 V and 4 V, respectively. What are the source rms voltage and average power supplied?
A) 10 V, 16 W
B) 22 V, 44 W
C) 10 V, 20 W
D) 14 V, 28 W

Q59 An ideal diode and a resistor R form a half-wave rectifier driven by $v(t) = V_0 \sin \omega t$. There is no smoothing capacitor. What is the ratio of average power dissipated in R to the average power that the same sinusoidal source would dissipate in R if R were connected directly to it? Neglect source resistance and diode voltage drop.
A) $1/\sqrt{2}$
B) 1/2
C) 1/4
D) 1

Q60 An ideal interferometer uses light of vacuum wavelengths 600 nm and 1000 nm. For each colour the two interfering beams are coherent and have equal amplitudes. At a reference point the geometric optical paths are initially equal, but one beam has an additional phase of π for each colour, so both colours are dark there. A plane glass slab is inserted normally in one path, replacing an equal thickness of air without otherwise changing the geometry. Its refractive index is 1.50 for both colours. What is the smallest positive slab thickness that makes the reference point an interference maximum for both colours?

A) 0.6 µm
B) 1.2 µm
C) 1.8 µm
D) 3.0 µm

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / B Plant kingdom: bryophytes and pteridophytes; dominant generations

Answer: (B) X is a moss and Y is a fern.

Why: A moss has a dominant gametophyte and a dependent, attached sporophyte. In a fern the conspicuous, independent leafy plant is the sporophyte. Gamete versus spore production identifies the generations before the plants are assigned.

Why not the others:

(A) Reverses the characteristic dominance and dependence.

(C) Does not explain Y's dominant independent sporophyte.

(D) Does not explain X's dependent sporophyte.

Key point: Identify the generation by what it produces, then use its independence to distinguish the plant groups.

Q2 / D Anatomy of flowering plants: monocot and dicot stems; open and closed bundles

Answer: (D) Q can establish a vascular cambium ring; P normally cannot.

Why: Q has open bundles, and its intrafascicular cambium can join interfascicular cambium during typical dicot-stem secondary growth. P describes the usual monocot stem with scattered closed bundles. The prediction concerns ordinary secondary growth, not exceptional monocot growth mechanisms.

Why not the others:

(A) Reverses the cambium evidence.

(B) Typical roots have radial vascular arrangement, not the described conjoint stem bundles.

(C) Q explicitly contains cambium and is therefore open.

Key point: Bundle position and cambium presence are separate observations; use both.

Q3 / A Biomolecules: enzyme inhibition and effects of temperature

Answer: (A) Competitive inhibition is consistent with I, while H has lost functional enzyme.

Why: At low substrate, I slows the reaction, but excess substrate restores approximately the untreated maximum rate. That is the competitive-inhibition pattern. H remains near one-tenth of U at both conditions, consistent with persistent heat denaturation reducing active enzyme. These results support an interpretation rather than uniquely proving a molecular inhibitor mechanism.

Why not the others:

(B) Irreversible loss would not be rescued by excess substrate; H is not restored.

(C) I never exceeds U's maximum, and H's low maximum does not show improved affinity.

(D) Substrate excess does not rescue H.

Key point: Distinguish substrate rescue from recovery of the enzyme's maximum activity.

Q4 / C Principles of inheritance and variation: linkage, recombination and gamete probabilities

Answer: (C) 0.64%.

Given: 1000 test-cross offspring; ab occurs 80 times.

Formula/Principle: A test-cross reads out the heterozygote's gametes. The aabb offspring must receive ab from each parent.

Step 1: The abundant classes Ab and aB are parental; the recombination fraction is (80 + 80)/1000 = 0.16.

Step 2: Each parent makes ab gametes with probability 80/1000 = 0.08.

Step 3: Independent gamete union gives P(aabb) = 0.08 × 0.08 = 0.0064 = 0.64%.

Worked derivation:

$$P(aabb) = P(ab)^2 = (80/1000)^2 = 0.0064 = 0.64\%.$$

Why: A test cross reports gamete frequencies directly because the recessive tester contributes no dominant allele. The linked gamete ab must be transmitted by both parents, so its measured frequency is multiplied by itself rather than replaced by an independent-locus ratio.

Why not the others: (A) is ten times too large; (B) squares a parental-gamete frequency; (D) treats the linked two-locus outcome as a single-gene recessive event.

Key point: Read gamete frequencies before combining linked loci.

Q5 / B Excretory products and their elimination: ADH, urine concentration and solute excretion

Answer: (B) Solute output is unchanged and water permeability increases.

Why: Daily osmole excretion equals urine volume times osmolarity: before, 2.0 × 300 = 600 mOsm/day; after, 0.75 × 800 = 600 mOsm/day. The same solute leaves in less water. With filtration and the medullary gradient held constant, increased collecting-duct water permeability, consistent with increased ADH action, explains the observations.

Why not the others:

(A) Lower volume does not imply lower solute output, and decreased permeability predicts the opposite water response.

(C) Confuses concentration with daily solute output.

(D) Gets the output right but reverses the permeability response.

Key point: Compare excretion rates, not concentration alone; these data are consistent with ADH action but do not prove a unique hormone concentration.

Q6 / A Ecosystem: gross/net productivity and trophic energy budgets

Answer: (A) 4%.

Given: Plant gross production 2400; plant respiration 900; herbivore assimilation 180 and respiration 120, all in kJ m−2 day−1.

Formula/Principle: NPP = GPP − producer respiration; net secondary production = assimilation − consumer respiration.

Step 1: Plant NPP = 2400 − 900 = 1500 kJ m−2 day−1.

Step 2: New herbivore biomass stores 180 − 120 = 60 kJ m−2 day−1.

Step 3: The requested percentage is (60/1500) × 100 = 4%.

Worked derivation:

$$\text{NPP} = 2400 - 900 = 1500, \quad P_H = 180 - 120 = 60, \quad 100P_H/\text{NPP} = 4\%.$$

Why: Producer respiration and consumer respiration belong to different energy budgets and must each be subtracted at the appropriate stage. The requested denominator is plant net production, so using either gross production or herbivore assimilation would compare different biological quantities.

Why not the others: (B) uses assimilation/GPP; (C) uses assimilation/NPP before consumer respiration; (D) uses consumer production/GPP instead of NPP.

Key point: Match the numerator and denominator to their trophic budgets.

Q7 / D Cell cycle and cell division: chromosome number and DNA replication

Answer: (D) 12 chromosomes and 4C DNA.

Why: S phase duplicates each DNA molecule to create sister chromatids. It doubles DNA content but does not yet separate centromeres into daughter chromosomes. The cell remains diploid in G2.

Why not the others:

(A) Counts chromatids as separate chromosomes before centromere separation.

(B) Omits DNA replication.

(C) Increases chromosome count while failing to double DNA.

Key point: DNA quantity and chromosome number need not change together.

Q8 / C Sexual reproduction in flowering plants: chromosome contributions in double fertilisation

Answer: (C) Zygote 36 with 1:2 contributions; endosperm nucleus 48 with 1:1 contributions.

Why: The female produces an egg and each polar nucleus with 12 chromosomes. Male meiosis produces gametes with 24. Syngamy gives 12 + 24 = 36, with maternal:paternal contribution 12:24. Triple fusion gives 12 + 12 + 24 = 48, with contribution 24:24. The usual 2:1 endosperm ratio assumes parents of the same ploidy and cannot simply be transferred here.

Why not the others:

(A) Assumes both parents are diploid.

(B) Applies the usual 2:1 ratio despite the specified gamete counts.

(D) Confuses the two fusion events.

Key point: Count the chromosomes in each participating nucleus before using a memorised ploidy ratio.

Q9 / A Photosynthesis in higher plants: chemiosmosis and ATP synthesis

Answer: (A) Transient ATP synthesis is possible without light-dependent NADPH formation.

Why: The lumen has higher proton concentration than the outside. Proton movement down this gradient through ATP synthase can power phosphorylation outside the vesicle. The finite imposed gradient dissipates, so ATP production is transient. A gradient by itself does not supply the light-driven electron transfer that reduces NADP+.

Why not the others:

(B) Exchanges the roles of electron transfer and the proton gradient.

(C) Ignores the energy already stored in the imposed gradient.

(D) Reverses the downhill proton direction and invents light-driven reduction in darkness.

Key point: Light normally establishes the gradient, but an imposed gradient can temporarily power ATP synthesis.

Q10 / B Human health and disease: passive and active immunity

Answer: (B) P acts sooner; Q can induce memory after a delay.

Why: Ready-made antitoxin antibodies supply passive protection while they persist. A toxoid acts as antigen and stimulates the recipient's own adaptive response, including memory formation. This is a comparison of mechanisms, not a treatment recommendation.

Why not the others:

(A) Neither preparation supplies ready-made memory cells.

(C) Interchanges passive protection and active memory formation.

(D) A toxoid supplies antigen, not ready-made antibodies.

Key point: Receiving antibodies and generating one's own immune response are different processes.

Q11 / C Biotechnology: principles and processes; restriction digestion, PCR and electrophoresis

Answer: (C) One-half.

Given: The sample contains bands at 900, 600 and 300 bp; complete digestion and equal amplification are stipulated.

Formula/Principle: R contributes two cut fragments; r contributes one uncut fragment. Fluorescence is proportional to base-pair mass, not simply to the number of bands.

Step 1: All three bands imply an Rr sample under the stated assumptions.

Step 2: For one equally amplified copy of each allele, band masses are proportional to 900, 600 and 300.

Step 3: The 900 bp fraction is 900/(900 + 600 + 300) = 1/2.

Worked derivation:

$$f_{900} = \frac{900}{900 + 600 + 300} = \frac{1}{2}.$$

Why: Complete digestion rules out residual uncut R DNA, so the three bands establish the presence of both alleles. Equal amplification gives equal total base-pair mass per allele, even though one allele contributes two separate bands.

Why not the others: (A) gives every band equal mass; (B) is the 300 bp mass fraction; (D) does not conserve the equal 900 bp contribution from the two alleles.

Key point: Fluorescence fractions depend on DNA mass, not band count.

Q12 / D Chemical coordination and integration: insulin action and feedback

Answer: (D) A defect between binding and the S-activated step is supported.

Why: Normal surface binding excludes failure of insulin binding as the explanation. Rescue by S shows that uptake machinery downstream of the bypass can respond. The failure of insulin alone therefore localises a functional defect to the interval between reception and that bypassed step; it does not identify a particular protein.

Why not the others:

(A) Both basal uptake and the S response contradict total loss of uptake.

(B) Externally supplied insulin tests response rather than cellular insulin synthesis.

(C) Contradicts the binding assay.

Key point: Combine the binding measurement and bypass control; neither measurement alone localises the defect.

Q13 / B Principles of inheritance and variation; Hardy–Weinberg principle and conditional carrier risk

Answer: (B) 1/18.

Given: The first parent is an unaffected child of Aa × Aa; the partner is unaffected and q = 0.20.

Formula/Principle: Condition genotype probabilities on being unaffected before multiplying transmission risks.

Step 1: The first person's carrier probability is (1/2)/(3/4) = 2/3.

Step 2: In the population, p = 0.80, so carrier frequency 2pq = 0.32 and unaffected frequency $1 - q^2 = 0.96$. The partner's conditional carrier probability is 0.32/0.96 = 1/3.

Step 3: Two carriers have an affected child with probability 1/4. Overall risk = (2/3)(1/3)(1/4) = 1/18.

Worked derivation:

$$P(aa) = \frac{2}{3}\frac{2(0.8)(0.2)}{1-0.2^2}\frac{1}{4} = \frac{1}{18}.$$

Why: Unaffected status changes each potential parent's genotype distribution before the offspring risk is evaluated. The independent family origins allow the two conditional carrier probabilities to be multiplied, followed by the one-quarter risk from two carriers.

Why not the others: (A) drops the first conditioning factor; (C) uses a 1/2 family carrier risk; (D) omits the partner's carrier uncertainty.

Key point: Condition both parental risks before multiplying transmission probabilities.

Q14 / A Structural organisation in animals: amphibian-frog; respiration

Answer: (A) Cutaneous diffusion fits the gradient-dependent uptake.

Why: Under the stipulated equal surface area, thickness and maintained blood flow, the oxygen concentration difference is the changing diffusion condition. P has twice Q's external oxygen and twice its measured uptake. Lowering P's external oxygen reduces that driving difference and should reduce uptake. This comparison supports skin exchange; it does not establish that it meets every possible metabolic demand.

Why not the others:

(B) Lung ventilation is excluded; increasing thickness would impede diffusion.

(C) Ignores the concentration gradient.

(D) Holding blood flow constant does not eliminate dependence on the external gradient.

Key point: Respiratory exchange depends on both surface properties and a maintained concentration difference.

Q15 / C Evolution: genetic drift, selection and population size

Answer: (C) Genetic drift is stronger in the small populations.

Why: With selection, migration and mutation excluded, finite sampling of parental alleles can change allele frequency. Opposite directions among replicate small populations, including loss and fixation, fit drift. Large populations show smaller sampling fluctuations.

Why not the others:

(A) Predicts a common selected direction and contradicts the controls.

(B) Infers a fitness advantage that the design excludes.

(D) Confuses genotype proportions under an ideal model with identical realised allele-frequency trajectories.

Key point: Replicate variability and population size can distinguish drift from a consistent selective advantage.

Chemistry

Q16 / D Kössel-Lewis structures, resonance and bond parameters

Answer: (D) Equivalent bonds with average bond order $4/3$.

Why: Each principal contributor contains one C=O bond and two C–O bonds. The resonance hybrid delocalises the bonding across equivalent oxygen positions; it is not a molecule switching between drawings. The model's average is $(2+1+1)/3 = 4/3$.

Why not the others:

(A) Incorrectly localises the double bond.

(B) Treats resonance contributors as separately interconverting molecules.

(C) Gives the model six bond-order units instead of four.

Key point: Resonance drawings are alternative representations of one delocalised structure.

Q17 / A Bonding in coordination compounds: ligand-field splitting and absorption

Answer: (A) Shorter wavelength for the cyanide complex; Co is +3 in both.

Why: In the stated transition model, $\Delta_o = h\nu = hc/\lambda$. A larger splitting therefore requires higher frequency and shorter wavelength. Ammonia is neutral. In the cyanide complex, $x + 6(-1) = -3$, giving $x = +3$.

Why not the others:

(B) Reverses the energy–wavelength relationship.

(C) Coordination number alone does not fix splitting.

(D) Mistakes the complex charge for the oxidation state of the central metal.

Key point: Determine metal oxidation state by charge balance, and spectral energy by ligand-field splitting.

Q18 / C Spontaneity and Gibbs energy change

Answer: (C) $+8.0$ and -8.0 $\mathrm{kJ\,mol^{-1}}$.

Given: $\Delta H = 24.0$ $\mathrm{kJ\,mol^{-1}}$; $\Delta S = 0.0800$ $\mathrm{kJ\,K^{-1}mol^{-1}}$.

Formula/Principle: $\Delta G = \Delta H - T\Delta S$.

Step 1: At 200 K, $\Delta G = 24.0 - 200(0.0800) = +8.0$ $\mathrm{kJ\,mol^{-1}}$.

Step 2: At 400 K, $\Delta G = 24.0 - 400(0.0800) = -8.0$ $\mathrm{kJ\,mol^{-1}}$.

Step 3: The sign changes at $T = \Delta H/\Delta S = 300$ K.

Worked derivation:

$$\Delta G(200) = 24 - 200(0.080) = 8, \quad \Delta G(400) = 24 - 400(0.080) = -8 \ \mathrm{kJ\,mol^{-1}}.$$

Why: The entropy contribution and enthalpy contribution must use the same energy units before subtraction. Positive entropy makes the free energy decrease with temperature, explaining why the two specified temperatures lie on opposite sides of the zero crossing.

Why not the others: (A) reverses the subtraction; (B) adds the entropy term; (D) mixes joules and kilojoules in the wrong direction.

Key point: Convert units and check the sign-changing temperature.

Q19 / B Chemical reactions: Tollens and iodoform tests; mixture analysis

Answer: (B) $1/2$.

Given: Identical portions are tested, so the inferred amounts can be compared directly.

Formula/Principle: One mole of aldehyde reduces two moles of silver ions; one mole of this methyl ketone gives one mole of iodoform.

Step 1: Silver amount is $2.16/108 = 0.0200$ mol, so propanal amount is $0.0200/2 = 0.0100$ mol.

Step 2: Propanone amount is $3.94/394 = 0.0100$ mol.

Step 3: $x_{\text{propanone}} = 0.0100/(0.0100 + 0.0100) = 1/2$.

Worked derivation:

$$n_{\text{aldehyde}} = \frac{2.16}{2(108)} = 0.0100, \quad n_{\text{ketone}} = \frac{3.94}{394} = 0.0100, \quad x_{\text{ketone}} = \frac{1}{2}.$$

Why: The two complete tests count different components because propanal supplies the Tollens response while propanone supplies the stipulated iodoform response. Identical portions make the resulting mole amounts directly comparable, but the two tests have different stoichiometric factors.

Why not the others: (A) treats one silver atom as one aldehyde molecule; (C) introduces an extra factor of two in the ketone amount; (D) divides the ketone fraction by two without a stoichiometric basis.

Key point: Convert each test response to component moles before calculating composition.

Q20 / D Nernst equation coupled to solubility equilibrium

Answer: (D) 0.74 V.

Given: The copper-ion concentration is controlled by precipitation equilibrium, not fixed at 1 M.

Formula/Principle: $K_{sp} = [\text{Cu}^{2+}][\text{OH}^-]^2$; for $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Zn}^{2+} + \text{Cu}$, $n = 2$.

Step 1: $[\text{Cu}^{2+}] = 10^{-20}/(10^{-4})^2 = 10^{-12}$ M.

Step 2: $Q = [\text{Zn}^{2+}]/[\text{Cu}^{2+}] = 10^{12}$.

Step 3: $E = 1.10 - (0.060/2)\log(10^{12}) = 1.10 - 0.36 = 0.74$ V.

Worked derivation:

$$[\text{Cu}^{2+}] = 10^{-12}\ \text{M}, \quad Q = 10^{12}, \quad E = 1.10 - \frac{0.060}{2}(12) = 0.74\ \text{V}.$$

Why: The copper half-cell is not at standard concentration because the buffer and solid hydroxide impose a solubility equilibrium. Its small free copper-ion concentration raises the reaction quotient and lowers the forward cell potential through the Nernst term.

Why not the others: (A) reverses the Nernst sign; (B) assumes standard concentrations; (C) omits the two-electron divisor.

Key point: Determine equilibrium ion concentration before applying the Nernst equation.

Q21 / C Amines and diazonium salts: interpreting staged nitrogen evolution

Answer: (C) 0.010 mol methylamine and 0.020 mol aniline.

Why: Primary aliphatic diazonium intermediates from methylamine decompose in the cold, giving one mole of nitrogen per mole of amine. Aniline forms a diazonium salt stable at 0–5 °C; its nitrogen evolves on warming. The first plateau therefore measures 0.010 mol methylamine. The additional gas is the difference between cumulative plateaus, 0.030 − 0.010 = 0.020 mol, and measures aniline.

Why not the others:

(A) Treats the final cumulative amount as methylamine.

(B) Reverses the two amines.

(D) Treats the final cumulative amount as additional warm-stage gas.

Key point: Use the chemical stability distinction and subtract cumulative readings before assigning the second component.

Q22 / A Oxidation numbers and charge balance in mixed-valence compounds

Answer: (A) $2/3$.

Why: Four oxide ions contribute charge −8. If x of the three iron ions are Fe(III), the others are Fe(II), so $3x + 2(3 - x) = 8$. Thus $x = 2$, giving two Fe(III) ions out of three iron ions. The fractional average oxidation number does not imply that each individual iron ion has oxidation state 8/3.

Why not the others:

(B) Is the Fe(II) fraction.

(C) Does not supply total charge +8.

(D) Would supply +9.

Key point: An average oxidation number can represent a mixture of integer oxidation states.

Q23 / B Integrated first-order rate equation and fractional conversion

Answer: (B) Approximately 38.2 min additional time.

Given: The first interval leaves 75% of the starting reactant; the target leaves 25%.

Formula/Principle: For first order, $kt = \ln([R]_i/[R]_f)$.

Step 1: $k = \ln(1/0.75)/10 = 0.0288$ min^{-1}.

Step 2: The additional interval takes concentration from 0.75 to 0.25 of the original, a ratio of 3.

Step 3: $\Delta t = \ln 3/k = 1.099/0.0288 \approx 38.2$ min.

Worked derivation:

$$k = \frac{0.288}{10} = 0.0288 \text{ min}^{-1}, \quad \Delta t = \frac{\ln(0.75/0.25)}{k} = \frac{1.099}{0.0288} \approx 38.2 \text{ min}.$$

Why: First-order kinetics depends on the ratio of reactant concentrations across the particular time interval. The additional interval starts after the first ten minutes, so it uses the decrease from three quarters to one quarter rather than from the original amount.

Why not the others: (A) assumes constant-rate percentage loss; (C) is the total time from the beginning; (D) confuses a remaining-fraction ratio with a number of equal time intervals.

Key point: Distinguish elapsed time from the requested additional interval.

Q24 / C Carbonyl reduction and selective functional-group transformations; alcohol acylation

Answer: (C) $C_8H_{14}O_2$, with four possible configurational stereoisomers.

Why: The starting ketone has formula $C_6H_{10}O$ and one stereocentre at the carbon bearing the methyl branch. Ketone reduction adds two hydrogens to give $C_6H_{12}O$ and creates a second stereocentre at the former carbonyl carbon. Alcohol acylation replaces its hydroxyl hydrogen with $COCH_3$, a net addition of C_2H_2O, yielding $C_8H_{14}O_2$. There are two independent stereocentres and no internal symmetry, so all four configurations are distinct. The terminal alkene has two identical hydrogens on one carbon and supplies no E/Z pair.

Why not the others:

(A) Omits the two hydrogens gained in carbonyl reduction.

(B) Counts only one stereocentre, or only products from one starting enantiomer.

(D) Incorrectly doubles the count for E/Z isomerism of a terminal alkene.

Key point: Track elemental changes, newly created stereocentres and the eligibility for geometric isomerism separately.

Q25 / D Structural and geometrical isomerism: interpreting alkene connectivity

Answer: (D) P/Q are position isomers, P/R are chain isomers, and only Q has E/Z forms.

Why: P is but-1-ene and Q is but-2-ene: the same unbranched carbon skeleton has the double bond at different positions. R is 2-methylpropene and has a branched skeleton, so it is a chain isomer of P. In Q, each double-bond carbon bears H and methyl, allowing geometrical isomerism. P and R each have a terminal CH2 group with two identical substituents on one double-bond carbon, preventing E/Z isomerism. All three have molecular formula C4H8.

Why not the others:

(A) Reverses the structural categories and counts ineligible terminal alkenes.

(B) Incorrectly grants E/Z isomerism to P.

(C) Reverses both the structural categories and E/Z eligibility.

Key point: Read the carbon skeleton before locating the double bond, then check both ends of the double bond for E/Z eligibility.

Q26 / B Osmotic pressure, dissociation and the van't Hoff factor

Answer: (B) Left to right; extra pressure $0.050RT$ atm on the right.

Given: $\alpha = 0.75$ for a salt yielding three ions on full dissociation.

Formula/Principle: $i = 1 + (3-1)\alpha$ and $\Pi = iCRT$.

Step 1: $i = 1{+}2(0.75) = 2.50$. Effective particle concentration on the right is $2.50(0.060) = 0.150$ mol L^{-1}.

Step 2: The left has 0.100 M particles, so water moves to the right.

Step 3: $\Delta\Pi = (0.150 - 0.100)RT = 0.050RT$ atm, applied on the right.

Worked derivation:

$$i = 1 + 2(0.75) = 2.5, \quad \Delta\Pi = [2.5(0.060) - 0.100]RT = 0.050RT.$$

Why: Water initially moves toward the higher effective solute-particle concentration because hydrostatic pressures begin equal. Partial dissociation changes that particle concentration, and a compensating pressure must be applied on the side that otherwise gains water.

Why not the others: (A) compares formula-unit concentrations only; (C) assumes full dissociation; (D) substitutes mass concentration for particle concentration.

Key point: Compare osmotic particle concentrations and the opposing pressure.

Q27 / A Mole concept and stoichiometry: decomposition followed by neutralisation

Answer: (A) 44.2%.

Given: Initial mass 1.90 g; final mass 1.59 g; lost mass 0.31 g.

Formula/Principle: $2\,NaHCO_3 \longrightarrow Na_2CO_3 + CO_2 + H_2O$. For 168 g bicarbonate, the volatile mass is 44 + 18 = 62 g.

Step 1: Initial bicarbonate mass is $0.31(168/62) = 0.84$ g.

Step 2: Its original mass percentage is $100(0.84/1.90) = 44.2\%$, to three significant figures. The remaining 1.06 g carbonate plus 0.53 g newly formed carbonate gives the observed 1.59 g residue.

Worked derivation:

$$m_{NaHCO_3} = 0.31\frac{168}{62} = 0.84\ \text{g}, \quad 100\frac{0.84}{1.90} = 44.2\%.$$

Why: Only the bicarbonate releases volatile products under the specified heating reaction, so mass loss measures that component selectively. Some bicarbonate mass remains as carbonate, which is why the lost mass cannot be equated directly to the original bicarbonate mass.

Why not the others: (B) loses the coefficient two in bicarbonate decomposition; (C) reports the original carbonate percentage; (D) treats all lost mass as bicarbonate.

Key point: Use the volatile mass per stoichiometric amount, then check the residue.

Q28 / D Electronic configurations and ionic-radius trends in isoelectronic species

Answer: (D) $S^{2-} > Cl^{-} > K^{+} > Ca^{2+}$ in ionic radius.

Why: Each ion has 18 electrons. With the electron count and occupied-shell pattern the same, the increasing nuclear charge pulls the electron cloud in more strongly. Nuclear charge rises from S to Ca, so radius decreases in that order.

Why not the others:

(A) Reverses the nuclear-charge trend.

(B) Does not follow the common-electron-count comparison.

(C) Reverses both the anion pair and the cation pair.

Key point: For an isoelectronic series, compare nuclear charge before applying neutral-atom size trends.

Q29 / C Carbohydrates: non-reducing disaccharides, hydrolysis and reducing sugars

Answer: (C) X sucrose, Y maltose, Z lactose.

Why: Sucrose is non-reducing and hydrolyses to glucose and fructose. Maltose is reducing and gives two glucose molecules. Lactose is reducing and gives glucose and galactose. X is therefore distinguished both by its initial test and its products; the identities of Y and Z follow from their different constituent monosaccharides.

Why not the others:

(A) Assigns a negative test to reducing maltose and glucose-only hydrolysis to sucrose.

(B) Interchanges the glucose-only and glucose/galactose disaccharides.

(D) Assigns the wrong reducing behaviour and products to X and Z.

Key point: Combine the intact molecule's reducing behaviour with the identities of its hydrolysis products.

Q30 / A Ionic equilibrium: neutralisation and a weak-base/conjugate-acid buffer

Answer: (A) Approximately 8.52.

Given: Initial ammonia amount $0.100(0.40) = 0.040$ mol; first HCl amount $0.100(0.10) = 0.010$ mol.

Formula/Principle: Complete neutralisation first, then evaluate the remaining weak-base buffer:

$$pOH = pK_b + \log_{10} \frac{[NH_4^{+}]}{[NH_3]}.$$

Step 1: After the first addition, 0.030 mol ammonia remains and 0.010 mol ammonium is present.

Step 2: The next 0.020 mol HCl leaves 0.010 mol ammonia and produces a total 0.030 mol ammonium; no strong acid remains in excess.

Step 3: Both buffer species share the same final volume, so their concentration ratio is $0.030/0.010 = 3$.

Step 4: $pOH = 5.000 + 0.477 = 5.477$; $pH = 14.000 - 5.477 = 8.523$.

Worked derivation:

$$n_{NH_3} = 0.010, \quad n_{NH_4^{+}} = 0.030, \quad pH = 14 - [5 + \log 3] = 8.523.$$

Why: The strong acid first changes the amounts of the weak base and its conjugate acid by stoichiometric neutralisation. Only after both additions are complete is the remaining ammonia/ammonium pair used as a buffer; their shared final volume cancels in the ratio.

Why not the others: (B) uses the buffer ratio before the second addition; (C) mistakes pKb for pH; (D) assumes partial neutralisation automatically gives a neutral solution.

Key point: Complete neutralisation bookkeeping before the buffer calculation.

Mathematics

Q31 / C Adjoint and inverse of a matrix

Answer: (C) 2.

Given: $A = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$.

Formula/Principle: $A^{-1} = \text{adj}(A)/\det A$.

Step 1: $\det A = 3(1) - 1(1) = 2 \neq 0$.

Step 2: $A^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 3 \end{pmatrix}$.

Step 3: The diagonal sum is $1/2 + 3/2 = 2$.

Worked derivation:

$$\det A = 2, \quad A^{-1} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 3 \end{pmatrix}, \quad \text{tr}(A^{-1}) = 2.$$

Why: The nonzero determinant permits the adjugate formula for the inverse. Its diagonal entries must be read from the inverse matrix, since inversion does not preserve the original trace.

Why not the others: (A) uses the original trace; (B) reports only the inverse-determinant factor; (D) introduces an unwarranted negative sign.

Key point: Check invertibility before evaluating an inverse trace.

Q32 / B Permutations and Combinations: constrained arrangements

Answer: (B) 96.

Given: Six positions, with B exactly two positions to the right of A.

Formula/Principle: Choose the allowed positions of the constrained pair, then permute all remaining distinct folders.

Step 1: A can occupy position 1, 2, 3 or 4; B then occupies 3, 4, 5 or 6, respectively. There are four ordered position pairs.

Step 2: The other four folders occupy the remaining positions in $4! = 24$ ways.

Step 3: The total is $4 \times 24 = 96$.

Worked derivation:

$$N = 4 \cdot 4! = 4(24) = 96.$$

Why: The left-to-right condition fixes the orientation of the constrained pair, so reversing it is not an additional allowed case. Once their two positions are fixed, all remaining folders are distinct and can occupy the four vacant places independently of which folder lies between A and B.

Why not the others: (A) loses half the allowed position pairs; (C) incorrectly allows B to the left of A as well; (D) treats A and B as a single adjacent block, which they are not.

Key point: Count allowed ordered positions before permuting the remainder.

Q33 / D Relations & functions: functional equations combined with minimisation

Answer: (D) $-2/3$.

Given: $f(x) + 2f(1-x) = x^2$ for all real x.

Formula/Principle: Use the equation at x and at $1-x$ to eliminate the second function value.

Step 1: Substitution gives $f(1-x) + 2f(x) = (1-x)^2$.

Step 2: Twice this new equation minus the original gives $3f(x) = 2(1-x)^2 - x^2 = x^2 - 4x + 2$.

Step 3: Hence $f(x) = ((x-2)^2 - 2)/3$.

Step 4: The point x = 2 belongs to the closed interval and makes the square zero, so the minimum is $-2/3$.

Worked derivation:

$$\begin{aligned} 3f(x) &= 2(1-x)^2 - x^2 \\ &= (x-2)^2 - 2, \\ \min_{[-1,2]} f &= -2/3. \end{aligned}$$

Why: The functional equation couples two values of the same function, so substituting the reflected argument creates a solvable pair of linear equations. Completing the square then identifies the minimum, and the closed interval matters because its endpoint attains that minimum.

Why not the others: (A) assumes a nonnegative right-hand side forces nonnegative f; (B) loses the constant's sign; (C) loses a factor during elimination.

Key point: Eliminate the reflected function value before optimising.

Q34 / A Complex Numbers: geometric loci in the Argand plane

Answer: (A) -1.

Given: Write $z = x + iy$.

Formula/Principle: Equal distances from +1 and −1 define the perpendicular bisector of those points.

Step 1: $(x-1)^2 + y^2 = (x+1)^2 + y^2$ gives x = 0.

Step 2: The other condition becomes $(y-2)^2 = 5$, with roots $y = 2 + \sqrt{5}$ and $2 - \sqrt{5}$.

Step 3: Their product is $(2+\sqrt{5})(2-\sqrt{5}) = 4 - 5 = -1$.

Worked derivation:

$$x = 0, \quad (y-2)^2 = 5, \quad y_1 y_2 = 4 - 5 = -1.$$

Why: The first distance equality is a line constraint rather than a fixed-radius condition. Intersecting it with the shifted circle yields two signed imaginary parts, whose product must retain its sign.

Why not the others: (B) takes the magnitude of the signed product; (C) ignores the circle's vertical displacement; (D) multiplies only the shared centre coordinate.

Key point: Translate each complex modulus into its own geometric constraint.

Q35 / C Definite-integral symmetry and substitution

Answer: (C) $\pi^2/4$.

Given: The weight $g(x) = \sin x/(1 + \cos^2 x)$ satisfies $g(\pi - x) = g(x)$.

Formula/Principle: Reflection of the limits gives $2I = \pi \int_0^\pi g(x)\,dx$.

Step 1: Replacing x by $\pi - x$ yields $I = \int_0^\pi (\pi - x) g(x)\,dx$; add this to the original integral.

Step 2: With $u = \cos x$, $\int_0^\pi g(x)\,dx = \int_{-1}^1 du/(1+u^2) = \pi/2$.

Step 3: Thus $I = (\pi/2)(\pi/2) = \pi^2/4$.

Worked derivation:

$$2I = \pi \int_0^{\pi} \frac{\sin x}{1+\cos^2 x}\,dx$$
$$= \pi \int_{-1}^{1} \frac{du}{1+u^2} = \frac{\pi^2}{2}, \quad I = \frac{\pi^2}{4}.$$

Why: The reflected weight is unchanged, which lets two versions of the integral combine to remove the explicit factor x. The remaining integral has a direct cosine substitution, so reflection and substitution perform different necessary parts of the solution.

Why not the others: (A) evaluates only the weight integral; (B) omits the division by two from reflection; (D) divides by two twice.

Key point: Use symmetry to simplify a weight before substituting.

Q36 / D Cartesian equation of a plane; normal projection and reflection

Answer: (D) $\sqrt{2}$.

Given: A normal to the plane is $\mathbf{n} = (1,1,1)$.

Formula/Principle: Reflection in $\mathbf{n}\cdot\mathbf{r} = d$ is $\mathbf{r}' = \mathbf{r} - 2(\mathbf{n}\cdot\mathbf{r} - d)\mathbf{n}/|\mathbf{n}|^2$.

Step 1: Here $\mathbf{n}\cdot P - d = 6 - 3 = 3$ and $|\mathbf{n}|^2 = 3$.

Step 2: Therefore $P' = (1,2,3) - 2(1,1,1) = (-1,0,1)$.

Step 3: $|OP'| = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2}$.

Worked derivation:

$$P' = P - 2\frac{6-3}{3}(1,1,1) = (-1,0,1), \quad OP' = \sqrt{2}.$$

Why: Reflection requires moving twice the perpendicular displacement to the plane, while projection moves only once. The requested final distance is measured from the origin, so it is not the same as the reflection displacement or the original point's distance.

Why not the others: (A) uses the original point; (B) uses the projection foot (0,1,2), not the reflection; (C) uses twice the perpendicular plane distance, not the origin distance.

Key point: Distinguish original point, projection foot and reflected point.

Q37 / B Differentiability of absolute-value functions and one-sided derivatives

Answer: (B) -4.

Given: Only x = 1 and x = −1 can be cusp points.

Formula/Principle: A term $c|x - x_0|$ has a derivative jump of $2c$; differentiability requires the total jump to vanish.

Step 1: Near x = 1, $|x^2-1| = |x-1||x+1|$ has leading cusp $2|x-1|$. The other nonsmooth contribution there is $a|x-1|$, so $a = -2$.

Step 2: Similarly, the cusp at x = −1 has coefficient $2 + b$, so $b = -2$. Terms centred at the other point are smooth locally.

Step 3: $a + b = -4$.

Worked derivation:

$$2 + a = 0, \quad 2 + b = 0 \quad \Rightarrow \quad a + b = -4.$$

Why: Each absolute-value cusp must cancel locally at its own position. A smooth term at the other cusp cannot cancel the derivative jump here, so the two coefficients are determined separately before they are added.

Why not the others: (A) reinforces rather than cancels the cusps; (C) cancels just one cusp; (D) treats cancellation of the coefficients' sum as enough at two distinct locations.

Key point: Check differentiability at every distinct nonsmooth point.

Q38 / A Statistics: mean and population variance of combined data

Answer: (A) 9.

Given: Equal group sizes; means 2 and 6; variances 1 and 9.

Formula/Principle: Combined variance equals the average within-group variance plus the variance of group means about the combined mean, with weights proportional to group sizes.

Step 1: Combined mean is $\mu = (2+6)/2 = 4$.

Step 2: Within-group contribution is $(1+9)/2 = 5$.

Step 3: Between-group contribution is $((2-4)^2 + (6-4)^2)/2 = 4$. Total variance is $5+4=9$.

Worked derivation:

$$\sigma^2 = \frac{1+9}{2} + \frac{(2-4)^2 + (6-4)^2}{2} = 5 + 4 = 9.$$

Why: Pooling changes the centre about which every deviation is measured, so variability between the group means contributes in addition to within-group variability. Equal group sizes justify equal weights, and the population convention fixes the divisor.

Why not the others: (B) omits between-group variation; (C) adds both squared mean offsets without averaging; (D) fails to normalise the final variance by the combined population size correctly.

Key point: Include both within-group and between-group variation.

Q39 / D Limits of algebraic functions: rationalisation

Answer: (D) 2.

Given: Both square roots are defined in a neighbourhood of zero.

Formula/Principle: Multiply the numerator and denominator by the conjugate.

Step 1: The numerator product becomes $(1+3x) - (1-x) = 4x$.

Step 2: For nonzero x near zero, the quotient equals $4/(\sqrt{1+3x} + \sqrt{1-x})$.

Step 3: Letting x tend to zero gives $4/(1+1) = 2$.

Worked derivation:

$$\frac{\sqrt{1+3x} - \sqrt{1-x}}{x} = \frac{4}{\sqrt{1+3x} + \sqrt{1-x}} \longrightarrow 2.$$

Why: Direct substitution gives a zero-over-zero form rather than the limit. Conjugate multiplication produces a common factor x that can be cancelled for nearby nonzero inputs, after which continuity can be applied to the remaining denominator.

Why not the others: (A) omits the conjugate denominator; (B) loses a factor of two; (C) substitutes into the indeterminate numerator without treating the simultaneous zero denominator.

Key point: Simplify on a punctured neighbourhood before evaluating a limit.

Q40 / C Probability with permutations and inclusion-exclusion

Answer: (C) $3/16$.

Given: There are $6! = 720$ equally likely assignments.

Formula/Principle: Choose the two correct assignments, then require that none of the other four is correct.

Step 1: Choose the two correctly assigned keys in $\binom{6}{2} = 15$ ways.

Step 2: By inclusion-exclusion, the number of permutations of the remaining four keys with no fixed assignment is $D_4 = 4! - \binom{4}{1}3! + \binom{4}{2}2! - \binom{4}{3}1! + \binom{4}{4}0! = 24 - 24 + 12 - 4 + 1 = 9$.

Step 3: Probability $= 15(9)/720 = 3/16$.

Worked derivation:

$$D_4 = 24 - 24 + 12 - 4 + 1 = 9, \quad P = \frac{\binom{6}{2} D_4}{6!} = \frac{15(9)}{720} = \frac{3}{16}.$$

Why: Exactly two correct placements means all four remaining placements must be incorrect. Ordinary permutations of the remainder would count extra fixed points, so inclusion-exclusion is needed to enforce the exact count.

Why not the others: (A) counts only four-cycles, omitting the three ways to form two disjoint swaps; (B) counts only the choice of the correct pair, omitting arrangements of the other keys; (D) allows all 4! assignments of the remainder and hence counts assignments with extra correct keys, some repeatedly.

Key point: Separate selection of fixed points from derangement of the remainder.

Q41 / A Inverse trigonometric functions: addition formula and principal values

Answer: (A) $3\pi/4$.

Given: Each inverse tangent is in $(0, \pi/2)$, and each exceeds $\pi/4$.

Formula/Principle: $\tan(\alpha+\beta) = (\tan\alpha + \tan\beta)/(1 - \tan\alpha\tan\beta)$.

Step 1: The tangent of the sum is $(2+3)/(1-6) = -1$.

Step 2: The sum is greater than $\pi/2$ and less than π, so it lies in the second quadrant.

Step 3: The second-quadrant angle with tangent -1 is $3\pi/4$.

Worked derivation:

$$\tan(\alpha+\beta) = \frac{5}{-5} = -1, \quad \frac{\pi}{2} < \alpha+\beta < \pi \Rightarrow \alpha+\beta = \frac{3\pi}{4}.$$

Why: The tangent of an angle determines it only up to an integer multiple of pi. The principal inverse-tangent ranges locate the sum in the second quadrant, which selects the positive second-quadrant angle instead of a negative principal arctangent.

Why not the others: (B) ignores the sum's quadrant; (C) loses the sign of the tangent denominator; (D) adds tangent arguments without the denominator correction.

Key point: Use range information after an inverse-trigonometric addition formula.

Q42 / B Solution by separation of variables with an initial condition

Answer: (B) 3.

Given: $y' = y/[x(1+x)]$, with $y(1) = 2$.

Formula/Principle: Separate variables and use $1/[x(1+x)] = 1/x - 1/(1+x)$.

Step 1: $\int dy/y = \int (1/x - 1/(1+x))\,dx$, so $y = Cx/(1+x)$ on the positive solution branch.

Step 2: $2 = C/2$, hence C = 4.

Step 3: $y(3) = 4(3)/4 = 3$.

Worked derivation:

$$y = \frac{Cx}{1+x}, \quad 2 = \frac{C}{2} \Rightarrow C = 4, \quad y(3) = 3.$$

Why: The differential equation separates on the positive solution branch through the initial point. The initial value fixes the integration constant before the requested later value is evaluated, preventing the constant from being mistaken for a function value.

Why not the others: (A) loses a factor when imposing the initial condition; (C) omits the final denominator change; (D) mistakes the integration constant for the requested function value.

Key point: Impose an initial condition on the complete solution.

Q43 / C Quadratic Equations: location of real roots

Answer: (C) $a > 3$.

Given: $f(x) = x^2 - 2ax + a + 2$ is a monic quadratic; the required root order is $r_1 < 1 < r_2$.

Formula/Principle: An upward-opening quadratic is negative exactly between its two distinct real roots.

Step 1:

$$f(1) = 1 - 2a + a + 2 = 3 - a.$$

Step 2:

$$f(1) < 0 \iff 3 - a < 0 \iff a > 3.$$

Step 3: This negative value and $f(x) \to +\infty$ at both ends ensure two distinct real roots, one on each side of one.

Worked derivation:

$$r_1 < 1 < r_2 \iff f(1) < 0 \iff a > 3.$$

Why: The sign at the specified test point handles the ordering directly rather than checking reality alone. The inequality is strict, so a root at one must be excluded.

Why not the others: (A) includes the invalid boundary $a = 3$; (B) only checks distinct-real-root conditions; (D) reverses the sign required between roots.

Key point: Combine root location with the leading coefficient's sign.

Q44 / D Conic sections: focal chord of a parabola, tangents and triangle area

Answer: (D) $8\sqrt{2}$ square units.

Given: The focus is (1,0), so the chord is y = x − 1.

Formula/Principle: Parametrise the parabola by $(t^2, 2t)$; its tangent is $ty = x + t^2$. Triangle area is half of base times perpendicular height.

Step 1: Substituting the chord gives $t^2 - 2t - 1 = 0$, so $t_1 + t_2 = 2$, $t_1 t_2 = -1$, and $t_1 - t_2 = 2\sqrt{2}$.

Step 2: Subtracting the two tangent equations gives $y_T = t_1 + t_2 = 2$, then $x_T = t_1 t_2 = -1$.

Step 3: The endpoints differ by $4\sqrt{2}$ in both coordinates, so PQ = 8. The distance of T from $x - y - 1 = 0$ is $4/\sqrt{2} = 2\sqrt{2}$.

Step 4: Area $= \frac{1}{2}(8)(2\sqrt{2}) = 8\sqrt{2}$.

Worked derivation:

$$t_1 + t_2 = 2, \quad t_1 t_2 = -1, \quad T = (-1, 2),$$
$$PQ = 8, \quad h = 2\sqrt{2},$$
$$\text{Area} = \tfrac{1}{2} PQh = 8\sqrt{2}.$$

Why: The focal chord determines both parabola parameters, and their sum and product locate the meeting point of the two tangents. Triangle area then uses the actual chord length and perpendicular distance to its line, not coordinate-wise distances.

Why not the others: (A) halves the base or height again; (B) uses a horizontal or vertical separation as the perpendicular height; (C) omits the one-half in triangle area.

Key point: Link intersections, tangent geometry and perpendicular height in order.

Q45 / B Dot/cross products and area of a parallelogram

Answer: (B) $3\sqrt{3}/2$ square units.

Given: $|\mathbf{a}| = |\mathbf{b}| = 1$ and the included angle is 60°.

Formula/Principle: Parallelogram area is the magnitude of the cross product of its adjacent sides.

Step 1: Expand $(\mathbf{a} + 2\mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = -\mathbf{a} \times \mathbf{b} + 2\mathbf{b} \times \mathbf{a} = -3\mathbf{a} \times \mathbf{b}$.

Step 2: $|\mathbf{a} \times \mathbf{b}| = 1(1)\sin 60° = \sqrt{3}/2$.

Step 3: The requested area is $3\sqrt{3}/2$.

Worked derivation:

$$(\mathbf{a} + 2\mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = -3\mathbf{a} \times \mathbf{b}, \quad \mathcal{A} = 3\sin 60° = \frac{3\sqrt{3}}{2}.$$

Why: Bilinearity of the cross product accounts for the linear combinations forming the two sides. Its magnitude gives parallelogram area, so the negative orientation sign must disappear but no triangle factor of one half is introduced.

Why not the others: (A) omits the linear-combination factor; (C) uses sin 60° = 1; (D) calculates triangle area instead of parallelogram area.

Key point: Expand the ordered product before taking its magnitude.

Physics

Q46 / A Distance and displacement from a signed velocity-time graph

Answer: (A) Average speed 2 m/s; average velocity zero.

Given: The piecewise linear graph crosses v = 0 at t = 3 s.

Formula/Principle: Displacement is signed area under v(t); distance is the area under |v(t)|.

Step 1: From 0 to 2 s, the triangular area is 4 m. From 2 to 3 s, it is another 2 m.

Step 2: The negative-velocity intervals 3 to 4 s and 4 to 6 s contribute signed areas -2 m and -4 m.

Step 3: Displacement is zero; distance is 4 + 2 + 2 + 4 = 12 m. Divide each by 6 s to get 0 and 2 m/s.

Worked derivation:

$$\Delta x = 4 + 2 - 2 - 4 = 0, \quad s = 12\ \text{m}, \quad \bar{v} = 0, \quad \overline{\text{speed}} = 12/6 = 2\ \text{m s}^{-1}.$$

Why: Velocity can be negative while distance travelled continues to increase. Signed graph areas therefore give displacement, whereas adding their magnitudes gives distance; both averages use the same total time.

Why not the others: (B) interchanges speed and velocity; (C) misses area beside the central zero crossing; (D) treats negative velocity as positive when calculating displacement.

Key point: Keep signed displacement separate from total distance.

Q47 / C Dimensional consistency of an empirical pressure model

Answer: (C) $ML^{-2}T^{-2}$.

Given: The term bh is added to pressure, and h has length dimensions.

Formula/Principle: Additive terms in a physical equation must have the same dimensions.

Step 1:

$$[P] = [\text{force}]/[\text{area}] = MLT^{-2}/L^2 = ML^{-1}T^{-2}.$$

Step 2:

$$[b][h] = [P] \Rightarrow [b] = [P]/L = ML^{-2}T^{-2}.$$

Step 3: The other pressure term is consistent since $[\rho v^2] = (ML^{-3})(L^2T^{-2}) = ML^{-1}T^{-2}$.

Worked derivation:

$$[b] = ML^{-2}T^{-2}.$$

Why: Dimensional consistency applies to each additive term, not just to the equation as a whole. Dividing the pressure dimensions by the height dimensions isolates the coefficient b.

Why not the others: (A) leaves pressure dimensions without dividing by height; (B) gives acceleration dimensions and omits density; (D) gives density dimensions and omits acceleration.

Key point: Isolate a coefficient by dividing the dimensions of its complete term.

Q48 / B Continuity and Bernoulli equation in a horizontal pipe

Answer: (B) $(15/2)\rho v^2$.

Given: The radius decreases by a factor of two, while elevation remains constant.

Formula/Principle: $A_1 v_1 = A_2 v_2$ and $P + \rho v^2/2$ is constant along a streamline.

Step 1: Area decreases by a factor of four, so the narrow-section speed is 4v.

Step 2: $P_1 - P_2 = \rho((4v)^2 - v^2)/2 = (15/2)\rho v^2$.

Worked derivation:

$$v_2 = 4v, \quad P_1 - P_2 = \tfrac{1}{2}\rho[(4v)^2 - v^2] = \frac{15}{2}\rho v^2.$$

Why: Continuity relates speed to cross-sectional area, whose radius dependence is squared. Bernoulli's equation then compares kinetic terms that square the resulting speeds again; the equal elevations remove only the gravitational term.

Why not the others: (A) doubles speed instead of quadrupling it; (C) omits the upstream kinetic-energy term; (D) mixes the area ratio with the kinetic-energy expression.

Key point: Apply area continuity before pressure-energy balance.

Q49 / D Angular momentum conservation and actuator work

Answer: (D) $4\omega/3$ and $mR^2\omega^2/2$.

Given: No external axial torque, but the actuator may supply mechanical energy.

Formula/Principle: $I_i\omega = I_f\omega_f$; supplied work equals the change in total mechanical kinetic energy.

Step 1: $I_i = 3mR^2$ and $I_f = 2mR^2 + m(R/2)^2 = 9mR^2/4$.

Step 2: $\omega_f = (3/(9/4))\omega = 4\omega/3$.

Step 3: $K_i = 3mR^2\omega^2/2$, while $K_f = \frac{1}{2}(9mR^2/4)(4\omega/3)^2 = 2mR^2\omega^2$.

Step 4: Supplied work $K_f - K_i = mR^2\omega^2/2 > 0$.

Worked derivation:

$$\omega_f = \frac{3}{9/4}\omega = \frac{4}{3}\omega,$$

$$K_f - K_i = 2mR^2\omega^2 - \frac{3}{2}mR^2\omega^2 = \frac{1}{2}mR^2\omega^2.$$

Why: An internal actuator supplies energy without supplying an external axial torque, so angular momentum is conserved while kinetic energy can change. Moving the bead inward decreases the system's moment of inertia, producing both a higher angular speed and a positive energy increase.

Why not the others: (A) reverses the inertia ratio; (B) confuses conservation of angular momentum with conservation of kinetic energy; (C) assumes fixed angular speed without an external torque.

Key point: Conserve the quantity protected by symmetry, not every form of energy separately.

Q50 / A Heat-engine energy balance and the Carnot bound

Answer: (A) Consistent with an irreversible engine.

Given: $Q_h = 800$ J, $W = 300$ J, $T_h = 500$ K and $T_c = 300$ K; the engine is cyclic.

Formula/Principle: Use the cyclic first law and the Carnot efficiency bound separately.

Step 1:

$$Q_c = Q_h - W = 800 - 300 = 500 \text{ J}.$$

Step 2:

$$\eta = W/Q_h = 300/800 = 0.375.$$

Step 3:

$$\eta_{\text{Carnot}} = 1 - T_c/T_h = 1 - 300/500 = 0.40.$$

Step 4: The first-law balance holds and $0.375 < 0.40$, so the values are compatible with an irreversible engine.

Worked derivation:

$$Q_h = W + Q_c, \qquad \eta = 0.375 < \eta_{\text{Carnot}} = 0.40.$$

Why: Conservation of energy fixes rejected heat but does not by itself decide reversibility. A reversible engine between these same reservoirs would reach the Carnot bound; the smaller stated efficiency is consistent with irreversibility.

Why not the others: (B) would require efficiency 0.40; (C) contradicts the verified energy balance; (D) reverses the numerical efficiency comparison.

Key point: Check both energy conservation and the temperature-dependent efficiency bound.

Q51 / B Spring force in vertical simple harmonic motion

Answer: (B) $3mg/2$.

Given: Equilibrium extension 0.10 m; initial displacement from equilibrium 0.03 m; initial speed divided by angular frequency 0.04 m.

Formula/Principle: SHM energy about equilibrium gives $A^2 = x_0^2 + (v_0/\omega)^2$, while $k(0.10) = mg$.

Step 1: $A = \sqrt{0.03^2 + 0.04^2} = 0.05$ m.

Step 2: The maximum extension from the natural length is $0.10 + 0.05 = 0.15$ m.

Step 3: The maximum spring tension is $k(0.15) = 3mg/2$. The minimum extension is 0.05 m, so the spring stays in tension.

Worked derivation:

$$A = \sqrt{0.03^2 + 0.04^2} = 0.05 \text{ m}, \quad T_{\text{max}} = k(0.10 + 0.05) = \frac{3}{2}mg.$$

Why: The spring force depends on extension from natural length, while the SHM displacement is measured from equilibrium. Initial kinetic and potential energies determine an amplitude larger than the initial displacement, and adding that amplitude to the static extension gives maximum tension.

Why not the others: (A) reports the maximum net restoring force about equilibrium; (C) adds displacement and velocity contributions linearly to overestimate the amplitude; (D) assumes equilibrium force balance holds throughout the motion.

Key point: Keep the natural-length and equilibrium reference positions distinct.

Q52 / C Resolving the ambiguity in a beat-frequency measurement

Answer: (C) 508 Hz and $(127/128)^2 T$.

Given: The initial string frequency is 512 Hz; the initial beat frequency is 4 Hz.

Formula/Principle: At fixed length and linear mass density, string frequency is proportional to the square root of tension. Beats equal the absolute frequency difference.

Step 1: The reduced string frequency is $512(255/256) = 510$ Hz.

Step 2: The first beat observation allows fork frequencies 508 or 516 Hz. Their differences from 510 Hz are 2 and 6 Hz, so only 508 fits both observations.

Step 3: For zero beats, the string must also be at 508 Hz: $T'/T = (508/512)^2 = (127/128)^2$.

Worked derivation:

$$f_s' = 512(255/256) = 510, \quad f_{\text{fork}} = 508 \text{ Hz}, \quad T'/T = (508/512)^2 = (127/128)^2.$$

Why: A single beat frequency leaves two possible fork frequencies on opposite sides of the string frequency. Changing tension moves the string frequency in a known direction, so the second observation resolves that ambiguity before the no-beat tension is calculated.

Why not the others: (A) subtracts a second beat difference from the fork estimate; (B) reports the changed string frequency; (D) uses only the first beat measurement.

Key point: Use a controlled frequency change to choose between beat-frequency branches.

Q53 / D Work from a nonuniform electric-field graph and a turning point

Answer: (D) $7a/4$.

Given: The electric force opposes the initial motion throughout the shown positive-x region.

Formula/Principle: $K(x) - K(0) = q \int_0^x E_x(s)\, ds$.

Step 1: From 0 to a, the signed triangular field area is $-E_0 a/2$, so $K(a) = qE_0 a(5/4 - 1/2) = 3qE_0 a/4$.

Step 2: Beyond a, the field is constant at $-E_0$. Stopping requires $qE_0(x - a) = 3qE_0 a/4$.

Step 3: Thus $x = a + 3a/4 = 7a/4$, inside the indicated interval.

Worked derivation:

$$K(a) = \frac{3}{4} qE_0 a, \quad qE_0(x - a) = K(a) \Rightarrow x = \frac{7}{4} a.$$

Why: The electric force does negative work throughout the forward motion, so kinetic energy must decrease continuously until the first zero. The graph changes form at a, requiring the triangular work before a to be accounted for before solving in the constant-field interval.

Why not the others: (A) uses the full initial energy with a constant field from x = 0; (B) miscounts the triangular area; (C) assumes the end of the drawn graph is a physical stopping boundary.

Key point: Accumulate signed work across each graph segment before locating a turning point.

Q54 / B An unbalanced resistor bridge with an ideal ammeter

Answer: (B) $12/17$ A from A to B.

Given: The ideal ammeter makes A and B equipotential, but it can carry a nonzero current.

Formula/Principle: Apply Ohm's law and Kirchhoff's current law after accounting for the ammeter connection.

Step 1: Above the common middle node, $R_u = 2 \parallel 3 = 6/5\,\Omega$. Below it, $R_l = 4 \parallel 3 = 12/7\,\Omega$.

Step 2: The supply current is $12/(6/5 + 12/7) = 70/17$ A. The middle-node potential relative to the bottom is $(70/17)(12/7) = 120/17$ V.

Step 3: Current arriving at A through 2 ohm is $(12 - 120/17)/2 = 42/17$ A; current leaving A through 4 ohm is $30/17$ A.

Step 4: The excess $12/17$ A flows from A through the meter to B.

Worked derivation:

$$V_{\text{mid}} = \frac{120}{17}\ \text{V}, \quad I_A^{\text{in}} = \frac{42}{17}, \quad I_A^{\text{out}} = \frac{30}{17}, \quad I_{A\to B} = \frac{12}{17}\ \text{A}.$$

Why: An ideal ammeter shorts the midpoint potentials but still permits current through that connection. The circuit must therefore be reduced with the midpoints joined; using disconnected voltage-divider results would describe a different circuit.

Why not the others: (A) uses the disconnected voltage-divider difference as a current; (C) reverses Kirchhoff's excess-current direction; (D) confuses zero meter voltage with zero meter current.

Key point: Rebuild the connected circuit before using Kirchhoff's current balance.

Q55 / A Mass number and atomic number in alpha and beta-minus decay

Answer: (A) Mass number 234 and atomic number 92.

Given: The parent has mass number 238 and atomic number 92; it emits one alpha particle and then two beta-minus particles.

Formula/Principle: Alpha emission changes (A, Z) by $(-4, -2)$; beta-minus emission changes it by $(0, +1)$.

Step 1:

$$(238, 92) \xrightarrow{\alpha} (234, 90).$$

Step 2:

$$(234, 90) \xrightarrow{\beta^-} (234, 91) \xrightarrow{\beta^-} (234, 92).$$

Worked derivation:

$$A_f = 238 - 4 = 234, \qquad Z_f = 92 - 2 + 1 + 1 = 92.$$

Why: An alpha particle removes two protons and two neutrons, while beta-minus conversion changes a neutron to a proton without changing nucleon number. Applying the changes in sequence prevents the beta steps from being mistaken for additional mass loss.

Why not the others: (B) stops after alpha emission; (C) counts beta changes but omits alpha; (D) removes only two nucleons in the alpha step.

Key point: Track mass and atomic numbers independently through a decay chain.

Q56 / D Motional emf and terminal speed on inclined rails

Answer: (D) $mgR\sin\theta/(B^2l^2)$.

Given: Field is normal to the plane of motion and the total resistance is constant.

Formula/Principle: Motional emf $\mathcal{E} = Blv$; current $I = Blv/R$; magnetic braking force $F_B = BIl$.

Step 1: $F_B = B^2l^2v/R$, directed up the incline by Lenz's law.

Step 2: At terminal speed the net force along the rails is zero: $mg\sin\theta = B^2l^2v_t/R$.

Step 3: Solve to obtain the stated expression.

Worked derivation:

$$\mathcal{E} = Blv, \quad F_B = \frac{B^2l^2v}{R}, \quad mg\sin\theta = F_B \Rightarrow v_t = \frac{mgR\sin\theta}{B^2l^2}.$$

Why: Motional emf creates a current whose magnetic force opposes sliding, so the braking force increases with speed. At terminal speed it balances the component of gravity along the incline; no kinetic-energy equality is needed because acceleration has become zero.

Why not the others: (A) uses the normal component of gravity; (B) inserts an inappropriate kinetic-energy factor of two into a force balance; (C) doubles the motional emf or gravitational component without justification.

Key point: Use induction to obtain the braking force, then apply terminal force balance.

Q57 / C Apparent depth through two plane liquid layers

Answer: (C) 7 cm; unchanged.

Given: Layer depths are 6 cm and 4 cm, with indices 3/2 and 4/3; the viewing medium is air.

Formula/Principle: In the paraxial approximation, apparent depth through parallel layers seen from air is $d_{\text{app}} = \sum_i t_i/n_i$.

Step 1: The contributions are $6/(3/2) = 4$ cm and $4/(4/3) = 3$ cm.

Step 2: Total apparent depth is 7 cm.

Step 3: Reversing layer order leaves the same two terms in the sum, so it does not change the result under the stated conditions.

Worked derivation:

$$d_{\text{app}} = \frac{6}{3/2} + \frac{4}{4/3} = 4 + 3 = 7 \text{ cm}.$$

Why: The near-normal approximation makes each parallel layer contribute its own depth divided by refractive index. Adding those contributions is independent of layer order when their depths and indices are held fixed, although more general oblique viewing would need a different treatment.

Why not the others: (A) ignores refraction; (B) incorrectly treats interface order as changing the paraxial sum for fixed layer depths; (D) multiplies depths by indices instead of dividing.

Key point: State the approximation before adding apparent-depth contributions.

Q58 / A Series RLC phasors and real power

Answer: (A) 10 V and 16 W.

Given: The reactive voltages are opposite in phase and the resistor voltage is in phase with current.

Formula/Principle: $V = \sqrt{V_R^2 + (V_L - V_C)^2}$; average power $P = IV_R$ for ideal L and C.

Step 1: Net reactive voltage is 10 - 4 = 6 V.

Step 2: Source voltage $= \sqrt{8^2 + 6^2} = 10$ V.

Step 3: Average power $= 2(8) = 16$ W; equivalently R = 4 ohm and $I^2R = 16$ W.

Worked derivation:

$$V = \sqrt{8^2 + (10-4)^2} = 10 \text{ V}, \qquad P = IV_R = 2(8) = 16 \text{ W}.$$

Why: Resistive voltage is in phase with current, while the net reactive voltage is perpendicular to it in the phasor diagram. Only the resistor dissipates average power for ideal reactive elements, so apparent power cannot replace the real-power calculation.

Why not the others: (B) adds all voltage magnitudes and equates apparent with real power; (C) uses $VI = 20$ VA as watts without a power factor; (D) combines the resistor and inductor voltages as in-phase scalars.

Key point: Add AC voltages as phasors and distinguish real from apparent power.

Q59 / B Half-wave rectification and mean power in a resistor

Answer: (B) $1/2$.

Given: The diode transmits one half-cycle unchanged and blocks the other.

Formula/Principle: Average resistor power is $\langle v^2 \rangle / R$, not $\langle v \rangle^2 / R$.

Step 1: For the direct sine wave, $\langle v^2 \rangle = V_0^2/2$, so $P_{\text{direct}} = V_0^2/(2R)$.

Step 2: Each half-cycle contributes equal squared-voltage area. Removing one gives $P_{\text{half}} = V_0^2/(4R)$.

Step 3: The power ratio is 1/2.

Worked derivation:

$$P_{\text{direct}} = \frac{V_0^2}{2R}, \quad P_{\text{half}} = \frac{V_0^2}{4R}, \quad \frac{P_{\text{half}}}{P_{\text{direct}}} = \frac{1}{2}.$$

Why: Heating depends on the time average of squared voltage, not the square of its mean. Removing one of two identical squared-sine half-cycle areas therefore halves the full-wave resistor power even though the surviving voltage peaks are unchanged.

Why not the others: (A) reports an rms-voltage ratio rather than a power ratio; (C) halves an already averaged voltage and squares it; (D) ignores that half the cycle carries no current.

Key point: Average instantaneous power over the full period.

Q60 / D Two-colour interference, phase reversal and minimum slab thickness

Answer: (D) $3.0\,\mu$m.

Given: Each colour begins with a relative phase of π; slab excess optical path is $\delta = (n-1)t = t/2$.

Formula/Principle: A bright fringe requires $\pi + 2\pi\delta/\lambda = 2\pi k$, so δ must be an odd multiple of $\lambda/2$ for each colour.

Step 1: We need $\delta = (2r+1)300\,\text{nm} = (2s+1)500\,\text{nm}$, with nonnegative integers r and s.

Step 2: The smallest common value with odd multipliers is $\delta = 1500$ nm: five times 300 and three times 500.

Step 3: $t = 1500/0.50 = 3000$ nm, or 3.0 micrometres.

Worked derivation:

$$\delta = 5(300) = 3(500) = 1500\ \text{nm}, \quad t = \frac{\delta}{n-1} = 3000\ \text{nm} = 3.0\ \mu\text{m}.$$

Why: The initial phase reversal makes brightness require an additional odd half-wavelength path difference for each colour. Finding the smallest path difference that satisfies both odd-multiple conditions is essential; satisfying one wavelength alone does not establish a shared maximum.

Why not the others: (A) satisfies only the 600 nm condition; (B) uses an integer-wavelength condition appropriate to no initial phase reversal and still misses the second colour; (C) again satisfies only the 600 nm odd-half-wavelength condition. The two different colours need not interfere with one another.

Key point: Retain initial phase when finding simultaneous interference maxima.
